NCCS
BSc CSIT First Sem: C-Programming
Lab assignment Set-1

1. Write a C program to find area and perimeter of rectangle.
2. Write a program to compute b^p. Your program should read value of base b and power p from user.
3. Write a program to compute floor and ceiling value of a real number.
4. Write a program to find simple interest and Amount.
5. Write a program to find surface area of a box. [S=2(lb+lh+bh)]
6. Write a program to find area of a triangle by reading its three sides. [$area = \sqrt{S(s-a)(s-b)(s-c)}$, where a, b, c are valid sides of triangle and S is its semi-perimeter]
7. Write a program to read temperature in Fahrenheit and convert it into centigrade. $\left(c = \frac{5(f-32)}{9}\right)$
8. Write a program to read temperature in centigrade and convert it into Fahrenheit.
9. Write a program to find area and circumference of circle using symbolic constant.[$area = \pi r^2$ $and\ Circumference = 2\pi r$]
10. Write a program to demonstrate the use of arithmetic operation on integers. (find sum, difference, product, quotient, and remainder).
    Write a program to read basic salary and calculate tax, allowance and net salary. Tax is 36% of basic salary and allowance is 25% of basic salary.
11. Write a program to find the sum of digits of three-digit number.
12. Write a program to read distance in meters and convert it into kilometers and meters. [1 km = 1000 m and output display output as **2575m=2 km and 575 m**]
13. Write a C program to read number of days and convert it into years, months and days.
14. Write a program to reverse four-digit numbers.
15. Write a program to check whether entered number is odd or even using ternary operator. **(? :)**
16. Write a program to read three different numbers and find the smallest one. (using ternary operator)

Programming on C
Lab assignment-set 2

1. Write a program to read a lowercase character and convert it into uppercase.
2. Write a program to read a number and check whether it is positive or negative.
3. WAP to check whether a number is odd or even.
4. WAP to read cost price and selling price of a good and find profit or loss amount.
5. WAP to read three numbers and find the largest one.
6. WAP to read three numbers and find the median number.
7. WAP a program to find the smallest among four given numbers.
8. Write a program to read a character and check whether it is letter, digits or others characters.
9. WAP a program to check whether a year is leap or not.
10. WAP to read coefficient of quadratic equation and finds all its roots.
11. WAP to read marks obtained in your final exams in all subjects and find total marks, percentage, results (pass or fail), division. (assume all the necessary condition)
12. WAP to read age of a person and display remarks accordingly:

Age	**Remarks**
Age<12	CHILD
12<=age<20	TEEN
20<=age<30	ADULT
30<=age<55	YOUNG
Age>=55	OLD

13. WAP to find sales person commission on sales amount according to following conditions

Sales amount	**Commission**
<5000	0%
>=5000 and < 10000	5%
>=10000 and <15000	7%
>=15000	10%

14. Write a menu base program for
    1. Area of rectangle
    2. Perimeter of rectangle
    3. exit
15. write a menu base program to perform calculation, according to following choice:
    + for addition
    - For subtraction
    * For multiplication
    / For division

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Programming on C
Lab assignment-set 3

Last date of submission:

1. Write a program to generate first **10** natural numbers using for loop.
2. Write a program to print your name 10 times.
3. Write a program to generate first **10** natural numbers using while loop.
4. Write a program to generate first **10** natural numbers using do-while loop.
5. Write a program to generate first '**n'** odd terms.
6. Write a program to generate first '**n'** even terms and find their sum also.
7. Write a program to find sum of first 'n' natural numbers.
8. Write a program to find the sum of square of first 'n' natural numbers. (using all loops)
9. Write a program to find sum of cube of first 10 natural numbers.
10. Write a program to display the first 10 terms of series 1, 5, 9, …
11. Write a program to find the factorial of a given number.
12. Write a program to find the sum of digits of a given number.
13. Write a program to find the reverse of a given number.
14. Write a program to check whether given number is prime or not.
15. Write a program to print all the prime numbers between 1 to 100.
16. Write a program to count total numbers of prime number between 1 to given number 'n'.
17. Write a program to display the first 'n' terms of Fibonacci series.
18. Write a program to display multiplication table up to 'n' as:

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30

19. Write a program to display the following output:
    A B C D E F G H I J
20. Write a program to display the following output:
    1
    1 2
    1 2 3
    1 2 3 4
    1 2 3 4 5
21. Write a program to display the following output:
    1 2 3 4 5
    1 2 3 4
    1 2 3
    1 2
    1
22. W rite a program to generate multiplication table up to number 'n' entered by user as:
    1×1=1
    1×2=2
    …
    1×10=10

23. Write a program to generate the following output:

```
* * * * *
 * * * *
  * * *
   * *
    *
```

24. Design the layout of chess board using nested loop. [hint use ASCII 219 for white and 255 for black square box]
25. Write a program to generate the pyramid of number as

```
        1
       121
      12321
     1234321
    123454321
```

26. Write a program to generate the following figure:

```
   *
  ***
 *****
*******
```

27. Write a program to generate the following figure:

```
   *
  ***
 *****
*******
 *****
  ***
   *
```

28. Write a program to generate the following figure:

```
ABCDEEDCBA
ABCD    DCBA
ABC        CBA
AB           BA
A              A
```

29. Write a program to generate the number as:

```
123454321
  1234321
   12321
    121
     1
```

30. Write a program to find the value of given base to the given power. [$y=a^b$]
31. Write a program to find the sum of following series:
    a) $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}$+… upto 10th terms
    b) 11+22+33+… upto n^{th} terms
    c) $1^1+2^2+3^3+4^4+$…. Upto n^{th} terms

Assignment-set-4

(Array and String)

1. Write a program to initialize 10 numbers in an array and display them.
2. Write a program to read N numbers from user and display the numbers, their sum and average.
3. Write a numbers to find largest and smallest number from set of N numbers entered by users.
4. Write a program to read N numbers and sort them in ascending order. (Using bubble sort)
5. Write a program to read N numbers and sort them in ascending order. (Using Selection sort)
6. Write a program to search an item in an array. (using linear search technique)
7. Write a program to search an item in an array. (using binary search technique)
8. Write a program to read N items and sort them in ascending or descending order depending upon user choice. (for e.g. 'a' for ascending and 'd' for descending)
9. Write a program to read N numbers in an array and find the sum of odd numbers.
10. Write a program to read N numbers in an array and then increment the elements by constant C, where C is entered by users.
11. Write a program to read age of 20 peoples and count the numbers of child, teen, adult, young, and old. Given
    - Age<13 -- child
    - Age>=13 and age<20 – TEEN
    - Age>=20 and age<31 – Adult
    - Age>=31 and age<55 – Young
    - Age>=55 – Old
12. Write a program to read N numbers and display only prime numbers.
13. Write a program to read salary of N employees and count the numbers of employee receiving salaries between 40000 to 60000.
14. Write a program to add two matrices by asking matrix order from user.
15. Write a program to find the sum of rows and column of matrix.
16. Write a program to multiply matrices a A and B.

    $$A = \begin{bmatrix} 1 & 5 & 2 \\ 3 & 4 & 6 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 1 \\ 2 & 3 \\ 5 & 2 \end{bmatrix}$$
17. Write a program to multiply two matrices by asking size from user.
18. Write a program to transpose a matrix.
19. Write a program to find sum of diagonal elements of square matrix.
20. Write a program to convert string into upper case without using function.
21. Write a program to find reverse of string without using function.
22. Write a program to copy string without using function.
23. Write a program to concatenate string without using function.
24. Write a program to check whether a word is palindrome or not.
25. Write a program to read line of text and count number of vowels, consonant, digits, white spaces and other characters.
26. Write a program to check whether a word is palindrome or not using function.
27. Write a program to remove all space from a line of text.
28. Write a program to count numbers of words from line of text.
29. Write a program to read name of N student and display in monitors.
30. Write a program to read N words and sort them in alphabetical order.
31. Write a program to sort {aman, akesh, anish, anam, anil, anju} in alphabetical order.

Assignment: 5

1. WAP to calculate simple interest using function.
    i. With function returning values and passing arguments
    ii. With function returning values and passing no argument
    iii. With function returning no values and passing arguments
    iv. With function returning no values and passing no arguments
2. WAP to find surface area of box using function.
3. WAP to find maximum of two number using function.
4. WAP a program to find smallest of three number using function.
5. Write a program to find factorial of number using function.
6. WAP to find HCF of two number using function.
7. Write a program to reverse a number using function.
8. WAP to check whether a number is Armstrong or not.
9. WAP to find roots of quadratic equation using function. Use function findroots() which receive three coefficient a,b,c of equation as arguments and display roots.
10. WAP a program to calculate base raised to the power by using function.
11. WAP that uses function **isprime()** for testing whether a number is prime or not.
12. Write a program using function to print line of 50 asterisk.
13. Write a program to display following triangle of asterisk of N lines using function.

```
   *
  ***
 *****
*******
```

14. WAP to find largest of N numbers using function.
15. WAP to find smallest of N numbers using function.
16. WAP to arrange list of numbers in ascending order using function.(using bubble sort or selection sort)
17. Write a program to add two matrices using user defined function. User read() function to read element of matrix, display() element of matrix in matrix form and add() function to add matrices.
18. Write a program to multiply two matrices using function.

19. WAP to find factorial of number using recursion.
20. WAP to find sum of first N natural number using recursion.
21. WAP to find product of two natural numbers using recursion.

22. WAP to find HCF of two numbers using recursive definition.
23. WAP to generate first n terms of Fibonacci series using recursive function to return N^{th} term of series.
24. WAP to calculate power of base using recursion.

Lab Assignment 6

1. Write a program to read bookid, title, author and price of 10 books using structure and display detail of those books whose author is "Dinesh Khadka".
2. Write a program to read roll, name and marks in 5 subjects of N numbers of students using structure and find total marks and percentage and finally display details in tabular format.
3. Write a program to read empid, name, post and salary of N employee using structure. And display detail of those employee whose salary is between 40000 and 60000.
4. Write a program to read empid, name, post and salary of N employee using structure. And arrange the records according to alphabetical order of name field.
5. Write a program to read roll, name, markc_in_C and marks_in_IIT of N students and then find average marks. Use pointer and DMA.
6. Write a program to read N numbers and find the largest and smallest using DMA and function.
7. Write a program to read N numbers using DMA and sort them in ascending order.
8. Write a program to add two matrices using DMA and function.
9. Write a program to reverse a string using user defined function. (do not use string.h)
10. Write a program to read line of text and count the number of vowels, consonants, digits, space and other characters using function. (pass address of variable storing result).
11. Write a program to copy a text file.
12. Write a program to read a positive integer and write its table in file "table.txt".
13. Write a program to write first N natural numbers in file "numbers.txt" and then read it and write all odd numbers in file "odd.txt" and even numbers in file "even.txt".
14. Write a program to write roll, name, marks in C of students in file "result.txt".
15. Write a program to set marks in C to 30 whose marks is less than 30 and greater than 27 in file "result.txt".
16. Write a program to delete record of a student from file "result.txt".
17. Write a program to replace all occurrence of words "Shiv" by "Vola" and "Krishna" by "Shyam" in file "sample.txt"
18. Write a program to draw two concentric circles of radii 70 and 100 respectively at the center of screen.
19. Write a program to draw square with each side 200 units and coordinate of top left corner (50,50).
20. Write a program to draw a triangle using line() function.